

THE PROBLEM

Attention is the scarce resource, not information

A population-health nurse working a panel between visits can read a chart quickly — trained reviewers clear six to twenty a day. **Ninety-four of these hundred patients have at least one abnormal lab.** Reading was never the bottleneck. What each finding costs *after* you read it is — the call, the order, the chase — and so is deciding which ones earn it.

The product answers one question: of these hundred, which twelve need you this week — and what to do about each. It is a ranked worklist, not a longer list of alerts.

And you can tell it what this week is about, in a sentence. “A diabetes clinic Thursday — glycaemic control and kidney screening, skip lipids” becomes a different brief for each specialist: HbA1c mentioned four times against zero in an unsteered run, statins zero against fifteen. A fixed scoring rule cannot be told that. The floor still fires and the cap still holds — it steers emphasis, never past the safety net.

74

lab orders never resulted

Oldest twelve months. Absence raises no alert anywhere, so nobody notices.

25 of 28

diabetics with no statin

Invisible one chart at a time. Only a systematic sweep finds it.

39 of 100

hold an impossible lab value

So any list built naively is partly built on records that cannot be correct.

Panel review · synthetic EHR, 100 patients, 153 annual wellness visits · real caseloads are multiples of this

WHAT THE DATA ACTUALLY IS

Four facts about this data, and what each one rules out

Every design decision below follows from a measured property of the extract, not from a preference. Each figure is a query in `demo_queries.sql`.

Finding	Evidence	What the product does about it
The notes are generated from the tables	Across 153 notes, zero diagnoses and zero medications appear that are not already rows. Strip the conditions, drugs, BP and BMI and what is left is four fixed sentences — one per template, no variation.	No extraction: there is no clinical fact in the prose to extract. But those four sentences assert that care was delivered — “labs ordered per guideline intervals” — and no column records that. So the notes became a check on the brief, and it finds three encounters where the note claims labs were ordered and not one ever resulted.
The two lab tables are different datasets	588 orders placed at the visit against 1,288 results spanning -727 to +345 days — only 127 of them within two months of it. No ORDER_PROC_ID is shared. Joining on patient-plus-analyte instead falsely resolves 85 of the 120 pending orders that triage narrows to 74.	Kept separate, each keyed to the encounter. “Never resulted” is read from a single row, never from a join — the only reason it is a finding rather than a fabrication. Match them on patient and analyte and a pending order acquires a result from a different year.
No medication is ever stopped	END_DATE and DISCON_TIME 100% null; all 522 read Active	“Patient is on X” is unanswerable. “Patient never had X” still holds — which is why care gaps survive as findings and duplicate-therapy claims do not.
Systolic and diastolic do not co-vary	Correlated at -0.37 where real pressures run +0.5 to +0.7, and pulse pressure is flat from 20 to 119 rather than peaked. 27 of 100 hold an impossible one; one reads 108/111.	Blood pressure is staged against ACC/AHA thresholds, and implausible readings are separated out as data defects rather than clinical findings.

ICD-10 codes fragment across near-duplicates, so cohorts are built on the 3-character category. Hypertension is 46 patients across two codes, not 25 — grouping on the raw code halves it.

Provider attribution is random ($\chi^2/df = 0.94$), so nothing in the product infers anything from who authored a note. Some authors are credentialed DVM.

Every figure here is reproducible from `demo_queries.sql` and the test suite — and the judging of it is itself a product surface. **Let me show you it running.**

BUILDING ON A COMPONENT THAT IS NOT DETERMINISTIC

Six ways this could be wrong, and the control for each

A language model is right most of the time, which is the hard case. The unit of design is the **control that recomputes the claim before anyone sees it**. Ordered by what a mistake costs a patient.

Failure mode	Control
A finding names the wrong patient	On the roster, and matching its own evidence
The nurse is told nothing when there is something	Nine findings computed before the agents run — no nine, no run; and the report is checked back against them
A stated number is wrong	Every figure comes from a tool, no opportunity for hallucination
A finding is lost between agents	Typed rows in a shared store, never prose
The brief contradicts the chart	Reconciled against the note — three claim labs that never resulted
A note tells the system what to do	No tools, fixed schema — a planted instruction is reported, not obeyed

Enforced twice: **69 tests** re-derive every tool’s answer in independent SQL, and **eight invariants** run across five live runs. A control nobody has watched fail is not a control, so every runtime control ships with a fault that trips it.

Four of the six are runtime checks, and each has a planted fault in `prototype/faults.py` or `fixtures.py`; the other two are structural and cannot fail at runtime

SAFETY AND WHAT IS ACTUALLY VERIFIED

No model computes what code can compute

Not a rule against using the model — it is what makes the judgement left to it worth trusting.

What is left to the model — and worth it

- Whether a gap is real given the whole regimen
- Whether an unreturned test still matters
- Whether a rate is clinically credible
- What to raise first, and in what words

None is a calculation. The model judged a 16% PCSK9 rate a question about the population rather than a defect, read three notes as claiming care that never happened, and chose which twelve to raise first.

What is measured eight invariants, five live runs

Invariants holding	Verified facts surfaced	Run-to-run overlap	Guaranteed findings
8 of 8	5 of 6	0.90	9 every run

Underneath, **69 tests** re-derive every tool’s answer in independent SQL — two seconds, no model. The verified fact not surfaced every run is the missing stop dates, at four in five.

Residual risk — named, not mitigated What is not controlled, and what each needs.

No contraindication data exists. No allergy, intolerance or preference anywhere in the extract, so “25 of 28 diabetics have no statin” cannot tell a gap from a documented reason. Needs: that data.

The ranking is corroborated, not validated. Every control opposite verifies a *fact*; none verifies a *priority*. An independent score reaches a similar shortlist — evidence, not proof. Needs: 25 pairwise comparisons.

Nothing recommends a drug or a dose, and nothing is actioned without a human reading the evidence.

Human in the loop: every finding carries the evidence behind it, and data-quality flags render above clinical content